NETFLIX DATA ANALYSIS

Using Python, Pandas, NumPy, Matplotlib & Seaborn

PROJECT DOCUMENTATION REPORT

Data Analysis Using Python

Project Information	Details
Author	Gulivindala Sushmitha
Completion	6 / 6 Tasks Completed
Environment	Python / Jupyter Notebook / VS Code
Repository	GitHub – Netflix-data-analysis1

GitHub Repository: https://github.com/GulivindalaSushmitha/Netflix-data-analysis1

This document presents the work completed in the six-task Netflix data analysis project. The report follows a task-oriented documentation format covering data preparation, content analysis, country-level exploration, release trends, rating and genre analysis, and final business insights.

The analysis was carried out as a practical Python data-analysis workflow, moving from the raw Netflix dataset to cleaned data, analytical outputs, visualizations, and conclusions.

Executive Summary

This project is an end-to-end exploratory analysis of Netflix Movies and TV Shows data. The main purpose is to transform a raw dataset into useful information about Netflix content. The work combines data cleaning, exploratory analysis, grouping and aggregation, visualization, and interpretation of the results.

The project is organized into six practical tasks. Each task focuses on a specific analytical question and produces information that contributes to the overall understanding of Netflix's content library. The workflow was implemented using Python libraries commonly used for data analysis and visualization.

Project Objectives

- Prepare the Netflix dataset for reliable analysis by checking and handling data-quality issues.
- Study the distribution of Movies and TV Shows in the catalog.
- Examine how Netflix content is distributed across countries.
- Identify patterns in content release years and growth over time.
- Analyze content ratings and genres to understand audience-oriented content patterns.
- Combine the findings into a final business-oriented insight report.

Tools & Technologies

Technology	Purpose in the Project
Python	Main programming language used for the complete analysis workflow.
Pandas	Loading, cleaning, filtering, grouping, transformation and aggregation of tabular data.
NumPy	Numerical operations and supporting data-processing tasks.
Matplotlib	Creation of charts for trends, comparisons and distributions.
Seaborn	Statistical and comparative visualizations with clearer analytical presentation.
Jupyter Notebook	Interactive execution and documentation of analysis steps.
VS Code	Project development, notebook execution and source-code management.
GitHub	Repository hosting and presentation of the completed project files.

Task 1 — Data Cleaning and Preparation

Objective: Prepare the Netflix dataset so that the remaining analysis can be performed consistently and accurately.

- Load the Netflix dataset into a Pandas DataFrame.
- Inspect the columns, records, data types and overall structure of the dataset.
- Identify missing or incomplete values in important fields.
- Apply suitable cleaning and transformation steps before analysis.
- Check the prepared dataset and retain the fields required for the following tasks.
- Create a reliable analysis-ready version of the data.

Skills demonstrated: data inspection, missing-value handling, preprocessing, Pandas operations, data-quality checking.

Result: The dataset was organized into a cleaner form that could be used for content, country, release-year, rating, genre and business-level analysis.

Task 2 — Content Type Analysis

Objective: Compare Movies and TV Shows and understand the composition of Netflix's catalog.

- Separate the dataset into Movie and TV Show records.
- Calculate the number of titles belonging to each content type.
- Compare the two categories using grouped counts.
- Represent the comparison with a suitable visualization.
- Interpret the difference between Movies and TV Shows in the catalog.

Analytical focus: The task converts simple category counts into an understandable view of Netflix's content mix. This provides a foundation for later comparisons by year, country, rating and genre.

Skills demonstrated: categorical analysis, filtering, value counts, aggregation, comparison charts and interpretation.

Task 3 — Country-wise Content Analysis

Objective: Explore the geographic distribution of Netflix content and identify countries with strong representation.

- Examine the country-related information available in the dataset.
- Organize country information so that content contributions can be compared.
- Calculate content frequency by country.
- Identify the leading countries based on the available records.
- Create a visualization to make the country comparison easier to understand.
- Interpret the geographic pattern of the Netflix catalog.

Skills demonstrated: grouping, aggregation, categorical comparison, ranking and geographic insight generation.

Task 4 — Release Year and Content Trend Analysis

Objective: Study how Netflix content is distributed across release years and identify important time-based patterns.

- Analyze the release_year field of the dataset.
- Count titles across different release years.
- Examine how the volume of content changes over time.
- Compare trends using suitable line or bar visualizations.
- Observe periods with higher or lower content representation.
- Summarize the main time-based pattern visible in the dataset.

Analytical value: Release-year analysis helps show how the catalog represents content from different periods and provides a time-based perspective for understanding Netflix's content library.

Skills demonstrated: time-series-style grouping, frequency analysis, trend visualization and interpretation.

Task 5 — Content Rating and Genre Analysis

Objective: Examine ratings and genres to understand the types of content represented in the Netflix dataset.

- Analyze the distribution of content-rating categories.
- Identify frequently occurring genres and content categories.
- Compare rating patterns across Movies and TV Shows where applicable.
- Use charts to present rating and genre distributions clearly.
- Explore which content categories appear most frequently.
- Translate the observed patterns into audience-oriented insights.

Skills demonstrated: feature exploration, category analysis, visualization design, comparison and insight generation.

Task 6 — Netflix Business Insights Report

Objective: Bring together the results from the earlier tasks and present the most useful findings in a business-oriented form.

- Review the results obtained from the complete exploratory analysis.
- Combine important findings from content type, country, release year, rating and genre analysis.
- Identify major patterns and trends visible in the Netflix dataset.
- Use visual outputs to communicate the findings clearly.
- Convert analytical observations into concise business-oriented insights.
- Prepare the final project-level report based on the completed analysis.

Skills demonstrated: exploratory data analysis, visualization, insight communication, analytical summarization and business interpretation.

Overall Analysis Workflow

Stage	Work Completed
1. Data Preparation	Loaded, inspected and prepared the Netflix dataset for analysis.
2. Content Analysis	Compared Movies and TV Shows and examined the catalog composition.
3. Country Analysis	Grouped and compared content representation across countries.
4. Time Analysis	Studied release-year distribution and content trends over time.
5. Rating & Genre Analysis	Examined ratings and genre patterns using analytical visualizations.
6. Business Insights	Consolidated findings into a final interpretation of the Netflix catalog.

Key Learning Outcomes

- Gained practical experience in loading and preparing a real-world tabular dataset.
- Improved understanding of Pandas filtering, grouping, aggregation and transformation.
- Practiced creating meaningful visualizations with Matplotlib and Seaborn.
- Learned how to move from raw records to interpretable analytical findings.
- Developed the ability to compare categories and identify patterns in content data.
- Practiced presenting technical analysis as concise, business-friendly insights.
- Used VS Code, Jupyter Notebook and GitHub as part of a complete project workflow.

Conclusion

The Netflix Data Analysis project demonstrates a complete exploratory data-analysis workflow using Python. The six tasks collectively cover the major stages of practical data analysis: preparing the data, examining content categories, studying geographic and time-based patterns, analyzing ratings and genres, and converting the findings into business-oriented conclusions.

The completed work provides a structured understanding of the Netflix content dataset while also demonstrating practical skills in Python, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook and VS Code. The project is maintained in a GitHub repository for reference and presentation.

Project Repository

https://github.com/GulivindalaSushmitha/Netflix-data-analysis1

— End of Project Documentation —